Physiology & Behavior, Vol 20, pp 7–10 Pergamon Press and Brain Research Publ , 1978 Printed in the U S A

Prolonged Cortisol Elevation in the Infant Squirrel Monkey after Reunion with Mother[1]

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(Received 9 September 1977)

LEVINE, S , C L COE, W P SMOTHERMAN AND J N KAPLAN *Prolonged cortisol elevation in the infant squirrel monkey after reunion with mother* PHYSIOL BEHAV 20(1) 7–10, 1978 – Pituitary-adrenal response in mother and infant squirrel monkeys was assessed following either 30 min separation or 5, 15 and 30 min after a brief separation and reunion of mother and infant Basal (undisturbed) samples were also obtained The results revealed that there was an increment (28%) in plasma cortisol levels in the separation-reunion condition This change was slight however when compared to the changes in plasma cortisol following separation (120%) A second experiment assessed plasma cortisol levels in mothers and infants that were reunited following a 30 min separation Blood samples obtained 30 min after reunion indicated that the plasma cortisol levels in mothers had returned to basal levels, whereas the infants remained highly elevated during this period Thus the effects of reunion following a very brief separation show the operation of an inhibitory process which occurs as a result of the interaction between mother and infant Reunion following a longer separation appears to be more effective in reducing the pituitary-adrenal response in the mother than in the infant

Mother-infant Separation Plasma cortisol

THE ONTOGENY of attachment of the infant primate to its mother or other relevant caretaking figures has been studied extensively, using a variety of techniques [4, 9, 11] For the most part, however, attachment in primates has been measured by some aspect of either affiliative or rejection behaviors One of the most widely used methodologies in this field has been the use of brief or protracted mother-infant separation, measuring the consequences of separation on the behavior of the infant and its subsequent effect upon reunion of the mother and infant Changes in the infant's behavior upon separation have been extremely well described by a number of authors and consist of an initial high level of agitation, which if separation persists, may lead to a state that has been described as being similar to certain aspects of human depression [5,6]

In an attempt to assess the effects of separation and reunion upon mother and infant squirrel monkeys, we have been studying the hormonal response to separation and to separation with immediate reunion [1,10] Hill, McCormack and Mason [3] demonstrated that the presence of an attachment object, a surrogate, reduced the adrenocortical response of young (up to 6 months of age) macaques to an unfamiliar environment In our initial studies using the pituitary-adrenal response as a measure of attachment in the squirrel monkey, it has been demonstrated that both the mother and infant show an adrenocortical response following a 30-min period of separation [10] In view of the reports of behavioral agitation and disturbance following separation, it was not surprising that the pituitary-adrenal system should reflect this behavioral state What was surprising in the Mendoza *et al* [10] study was that when the mother and infant were briefly separated and then reunited, both failed to show an adrenocortical response There was no elevation in circulating levels of plasma cortisol compared to basal levels

In the Mendoza *et al* experiment mother and infant pairs were housed individually in dyads The effects of separation and separation-reunion have also been demon-

[1] This study was supported by Research Grant NICH&HD–02881 from the National Institutes of Health (to S L) The authors would like to thank Cathy McKenzie for her assistance

[2] Supported by USPHS Research Scientist Award K5–MH–19936 from the National Institute of Mental Health

[3] Supported by Biosciences Training Grant MH–8304

[4] Supported by the C A Aaron Endowment Fund

[5] Supported by Research Grant NICH and HD 04905 from the National Institutes of Health

strated in a subsequent study with group-living animals [1] In this latter study, there was an additional finding of interest Infants that remained in the group-living environment after their mother was removed, although aunted by a term pregnant female, also showed elevations in plasma cortisol It thus appeared that although there was no clear behavioral disturbance observed in the infants that were aunted, there was a definitive physiological response to separation

The present study was designed as an extension of the Mendoza *et al* report and used pair-housed animals In the previous study only one time point – 30 min following separation or separation-reunion – was obtained, and it is possible that there may have been a very rapid response to separation which was not observed because of the temporal relationship between the reunion after separation and the time that blood was sampled The purpose of the first experiment presented here was to establish a time course for pituitary-adrenal responsiveness to separation by obtaining blood samples from mothers and infants at selected intervals following separation

In a second experiment we examined the extent to which the stress response was reduced by reunion after high levels of plasma cortisol had already occurred following 30 min of separation

EXPERIMENT 1

Animals

The squirrel monkeys *(Saimiri sciureus)* used in this study were of the Colombian, Peruvian and Bolivian varieties All adults were wild born Experiment 1 consisted of 7 mother-infant pairs beginning when the infants were approximately 3 months of age All mother-infant pairs were housed in standard basket-type primate cages with food and water continuously available They were housed in a controlled environment with constant temperature and humidity with a 12 hr light/12 hr dark cycle, lights on at 0700 All testing occurred between 1500 and 1700 hr

Method

Each mother-infant pair was subjected to three experimental conditions (1) Base sampling Mother and infant were removed from their home cage, separated, immediately anesthetized and their blood was sampled (2) Separation Mother and infant were removed together from their cage and separated The mother was then returned to her cage and the infant placed in an identical holding cage in a different room for 30 min. Following this period mother and infant were anesthetized and blood sampled in their respective locations (3) Separation-reunion Mother and infant were removed, separated, immediately reunited and returned to their home cage Blood samples were obtained at different times following reunion 5 min, 15 min and 30 min

Each blood sample was obtained on a separate occasion One week was permitted to elapse between samples for each pair The conditions were ordered in a balanced design to control for any possible effects of repeated sampling No behavioral observations were obtained since the presence of an observer might have affected plasma cortisol levels

Plasma Cortisol Determinations

The technique for blood sampling and for radioimmunoassay of cortisol for the squirrel monkey has been described in previous publications [10] Blood was collected in heparinized syringes by cardiac puncture while animals were anesthetized following a brief exposure to ethyl ether All samples were rapidly collected within 3 min of catching the animal The blood was centrifuged, the plasma collected and frozen until assay Cortisol was assayed using antiserum #F2153 from Endocrine Sciences, Tarzana, CA, as described by Klemm and Gupta [7] Further details of these analyses have been presented in our previous publications [1,10]

RESULTS AND DISCUSSION

These data were analyzed using a 2 (Mother vs Infant) × 5 (Conditions) analysis of variance The results revealed that there was no significant difference between the response of mothers and infants There was a significant difference among conditions, $F(4/48) = 13.68$, $p<0\ 001$ Post hoc analyses (t-tests for correlated means) indicated there was a significant elevation of plasma cortisol at 5 min ($t = 2\ 74$, $p<0\ 02$) and 30 min ($t = 2\ 36$, $p<0\ 05$) in the separation-reunion condition but not at 15 min The plasma cortisol values in the separation condition were all significantly different from the separation-reunion values (Separation vs Separation-Reunion, 5 min, $t = 3\ 43$, $p<0\ 01$, Separation vs Separation-Reunion, 15 min $t = 5\ 27$, $p<0\ 01$, Separation vs Separation-Reunion, 30 min, $t = 3\ 05$, $p<0\ 01$) Thus, although the values obtained in the separation-reunion condition did differ significantly from basal values, the increase in plasma cortisol was only 28% In contrast, the increment in plasma cortisol values in the separation condition was 120% over basal values (see Fig 1)

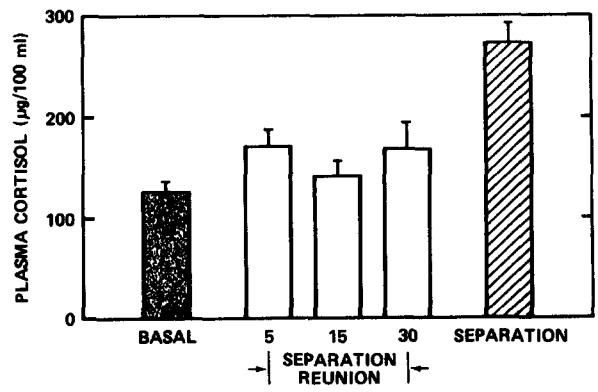


FIG 1 Mean (± SE) plasma cortisol values for each experimental condition

These results are consistent with those previously reported on the effects of separation and separation-reunion in mother and infant squirrel monkeys [1,10] The effects of separation on plasma cortisol have now been replicated in three experiments in our laboratory and clearly show that the pituitary-adrenal system is activated by this manipulation The separation-reunion condition had previously been tested only at the 30 min point and although the data from the present experiment are somewhat different from our previous results in that a statistically significant rise in plasma cortisol was found both at 5 min

and 30 min, this rise was in no way comparable to the elevation seen following a 30 min period of separation These data, therefore, support the hypothesis that the opportunity to interact with the focus of attachment suppresses or eliminates the pituitary-adrenal response following the disturbance of a relatively brief separation Further, they are consistent with the interpretation that one of the functions of the mother-infant relationship is to buffer both members of the dyad from environmental perturbations which could influence the pituitary-adrenal system and possibly result in behaviors detrimental to the mother-infant relationship [12]

EXPERIMENT 2

If reunion does function to buffer the stress response of the mother and infant to separation, the question remains as to whether this process would occur following a longer period of separation after elevated levels of plasma cortisol had already been attained In the following experiment mothers and infants were separated for 30 min, reunited, and blood samples were taken 30 min later to determine whether reunion would suppress the elevated plasma cortisol values which have been demonstrated to occur following a 30 min period of separation

Animals

The animals used were the 7 squirrel monkey mother-infant pairs which had been used in Experiment 1 and the data for the present experiment were collected during the same period as the first experiment

Method

Each mother-infant pair was subjected to three experimental conditions (1) Basal sampling, (2) Separation, and (3) Reunion The first two conditions and the values of plasma cortisol used for comparison purposes were those from Experiment 1 The third condition represented a procedural variation of the separation-reunion treatment In this condition the mother and infant were removed together from their home cage and separated for 30 min as described previously Following separation they were returned to their home cage and 30 min later sampled for blood Blood samples for each condition were obtained on separate occasions and 1 week was permitted to elapse between samples for a mother-infant pair A balanced design was again used to control for the possible effects of repeated sampling Blood samples and plasma cortisol determinations were completed as previously described

RESULTS AND DISCUSSION

A 2 (Mothers vs Infants) × 3 (Conditions) analysis of variance revealed a significant difference between the values for mothers and infants, $F(1/24) = 12\ 91$, $p<0\ 01$, a significant effect of Conditions, $F(2/24) = 20\ 67$, $p< 0\ 001$, and a significant Subjects × Conditions interaction, $F(2/24) = 8\ 38$, $p<0\ 01$ These results are represented in Fig 2 and indicate that reunion following the 30 min separation had differential effects on mothers and infants Post hoc analyses (t-tests for correlated means) indicated that the separation-induced plasma cortisol values for mothers were significantly higher than either the basal values ($t = 3\ 10$, $p<0\ 01$) or the Reunion values ($t = 2\ 90$, $p<0\ 05$), which did not differ from each other In contrast, the Separation

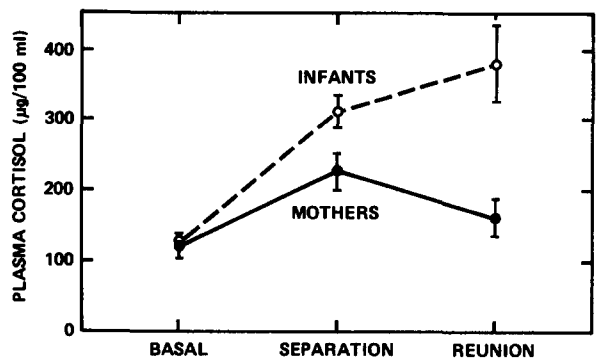


FIG 2 Mean (± SE) plasma cortisol concentrations for mothers and infants for each experimental condition

and Reunion values for infants were both significantly different from infant basal levels ($t = 8.18$, $p<0\ 01$, $t = 4\ 42$, $p<0\ 01$), and did not differ significantly from each other That is, infant levels did not return to base following reunion after separation

These latter data are open to several possible interpretations The first would suggest that the nature of the mother-infant interaction leads to differential effects on each member of the dyad The effects of reunion are more immediately observable in the mother following a 30 min period of separation It appears as though the mother's plasma cortisol returns to basal level during the period of time in which the infant is still showing a vigorous pituitary-adrenal response This interpretation has some appeal from an ethological point of view since it may be argued that the stability of the mother is essential in maintaining the mother-infant relationship If the mother were to remain disturbed following reunion her capacity to provide adequate care to the infant could indeed be impaired

An alternate hypothesis exists which could also account for these data It is known that the infant mammal's pituitary-adrenal system differs in many characteristics from that of the adult [8] One of the ways in which the infant pituitary-adrenal system differs, at least in the rat, is that there is a difference in the maturation of the negative feedback Thus, a brief stress to an adult animal leads to an initial activation of the pituitary-adrenal system which, if followed over a period of time, returns to basal values In contrast, it has been demonstrated that prepubertal animals, although they show the same pituitary-adrenal activation to a stimuli such as ether and electric shock, remain elevated for longer periods of time [2] It is therefore possible that we are dealing with differences between a mature and an infantile pituitary-adrenal system in the squirrel monkey, and that once activated the infant requires a considerably longer period of time before it returns to basal level in contrast to the fairly rapid return in the adult female Within the confines of the present experiment it is not possible to choose between these two obvious alternate hypotheses The parametric studies necessary to examine the time course of the influence of separation and separation-reunion, and, in addition, studies examining the maturation of the pituitary-adrenal system in the squirrel monkey have yet to be performed

As a whole, however, these data speak to the general processes of mother-infant attachment The effects of

reunion following brief separation clearly show the operation of an active inhibitory process which is a result of the interaction of mother and infant Reunion following a longer separation appears to be more effective in reducing the pituitary-adrenal activation in the mother than in the infant

REFERENCES

1 Coe, C L , S P Mendoza, W P Smotherman and S Levine Mother-infant attachment in the squirrel monkey Adrenal response to separation *Behav Biol* , in press

2 Goldman, L , C Winget, G W Hollingshead and S Levine Postweaning development of negative feedback in the pituitary-adrenal system of the rat *Neuroendocrinology* **12** 199–211, 1973

3 Hill, S D , S A McCormack and W A Mason Effects of artificial mothers and visual experience on adrenal responsiveness of infant monkeys *Devl Psychobiol* **6** 421–429, 1973

4 Hinde, R A and Y Spencer-Booth Effects of brief separation from mother on rhesus monkeys *Science* **3** 111–118, 1971

5 Jones, B C and D L Clark Mother-infant separation in squirrel monkeys living in a group *Devl Psychobiol* **6** 259–269, 1973

6 Kaufman, I C and L A Rosenblum The reaction to separation in infant monkeys Anaclitic depression and conservation withdrawal *Psychosom Med* **29**: 648–675, 1967

7 Klemm, W and D Gupta A routine method for the radioimmunoassay of plasma cortisol without chromatography In *Radioimmunoassay of Steroid Hormones,* edited by D Gupta Weinheim Verlag Chemie, 1975, pp 143–151

8 Levine, S and L J Treiman Role of hormones in programming the central nervous system In *Ciba Foundation Symposium on Foetal Autonomy,* edited by G E W Wolstenholme and M O'Connor London J and A Churchill, 1969, pp 271–281

9 Mason, W A Motivational factors in psychosocial development In *Nebraska Symposium on Motivation,* edited by W J Arnold and M M Page Lincoln University of Nebraska Press, 1971, pp 35–67

10 Mendoza, S P , W P Smotherman, M T Miner, J Kaplan and S Levine Pituitary-adrenal response to separation in mother and infant squirrel monkeys *Devl Psychobiol* , in press

11 Rosenblum, L A Infant attachment in monkeys In *The Origins of Human Social Relations,* edited by R Schaffer New York Academic Press, 1971, pp 85–113

12 Smotherman, W P , S G Wiener, S P Mendoza and S Levine Pituitary-adrenal responsiveness of rat mothers to noxious stimuli and stimuli produced by rat pups In *Breast-feeding and the Mother,* edited by K Elliott and D W Fitzsimons, Ciba Foundation Symposium 45 Amsterdam Elsevier, 1976, pp 5–25